ED-Explain: Personalized Video Instructions for Patients Discharged from the Emergency Department

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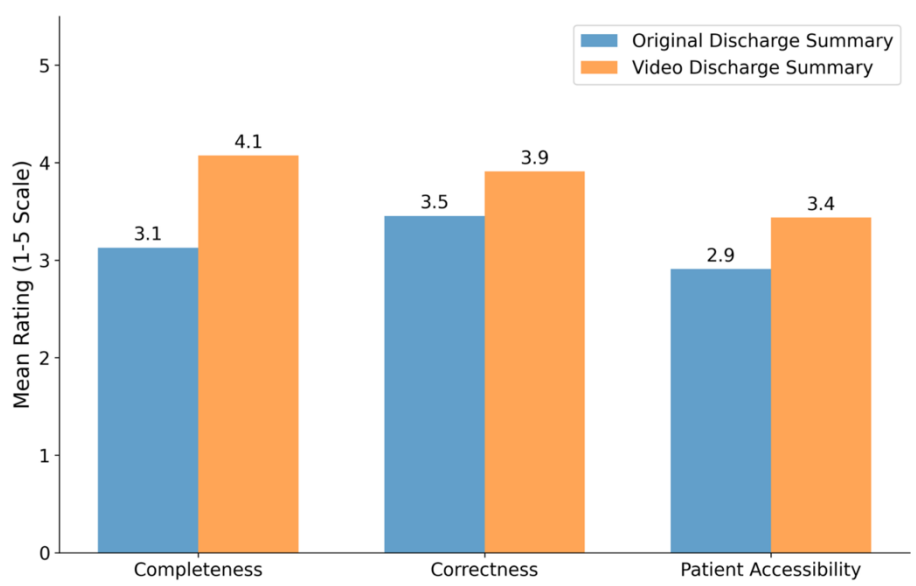
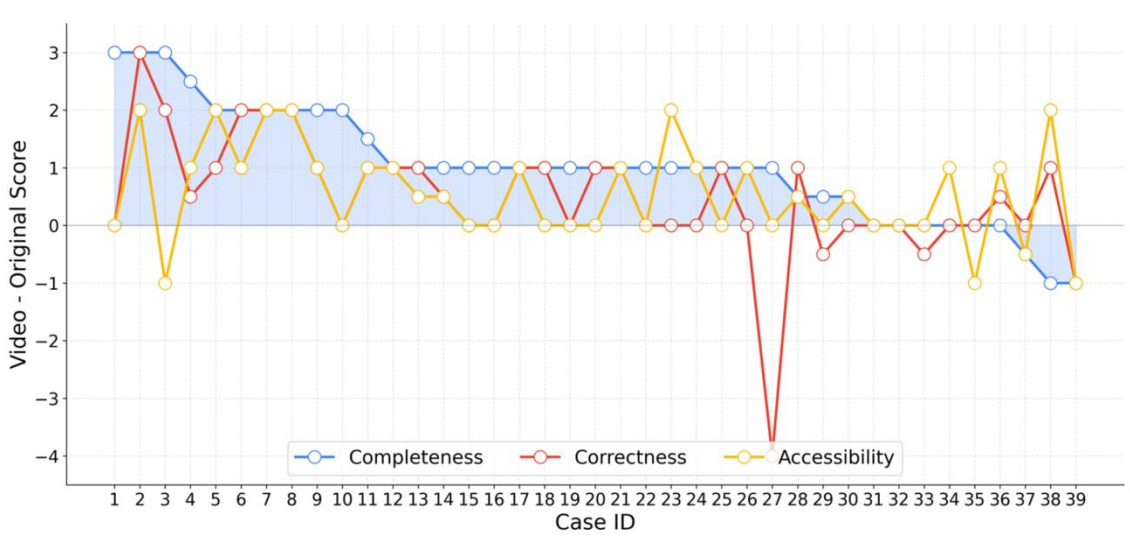
MOTIVATION & CHALLENGES

- **Communication Crisis:** 41-78% of ED patients fail to understand discharge instructions, leading to poor adherence and preventable return visits.
- **Health Literacy Barrier:** Nearly 9 in 10 adults struggle with complex medical terminology in health information.
- **Implementation Gap:** Existing solutions require excessive clinician time, making them impractical for high-volume ED settings.

RESULTS

- **Superior Completeness:** AI instructions rated 4.1 vs 3.1 for original ($p<0.001$), with 77% of cases showing improvement.
- **Enhanced Correctness:** AI scored 3.9 vs 3.5 for original ($p<0.001$), maintaining clinical accuracy while improving clarity.
- **Better Accessibility:** AI rated 3.4 vs 2.9 for original ($p<0.001$), achieving patient-friendly language without oversimplification.
- **High Acceptability:** Only 4.6% of AI videos deemed inappropriate for patients; 60% suitable for independent viewing.





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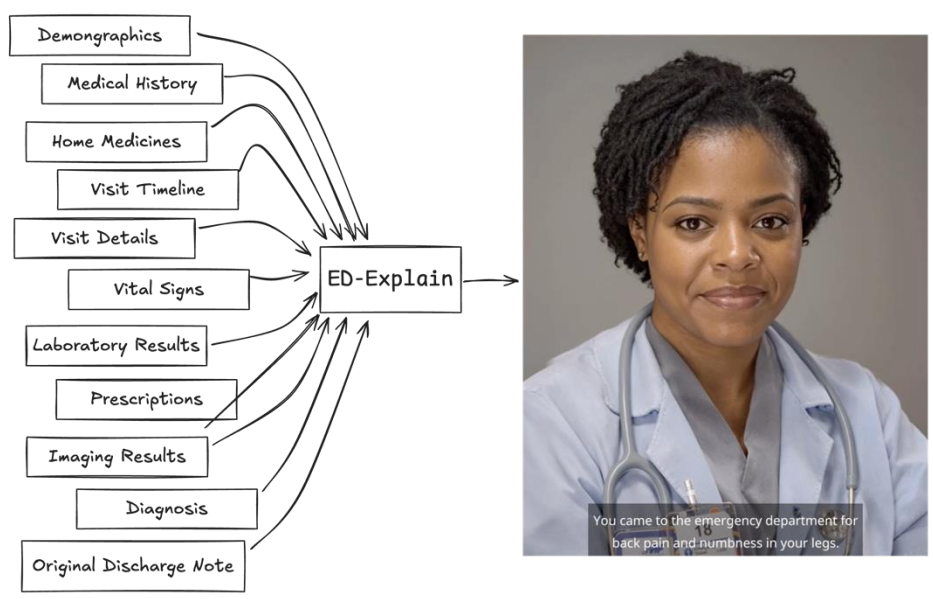
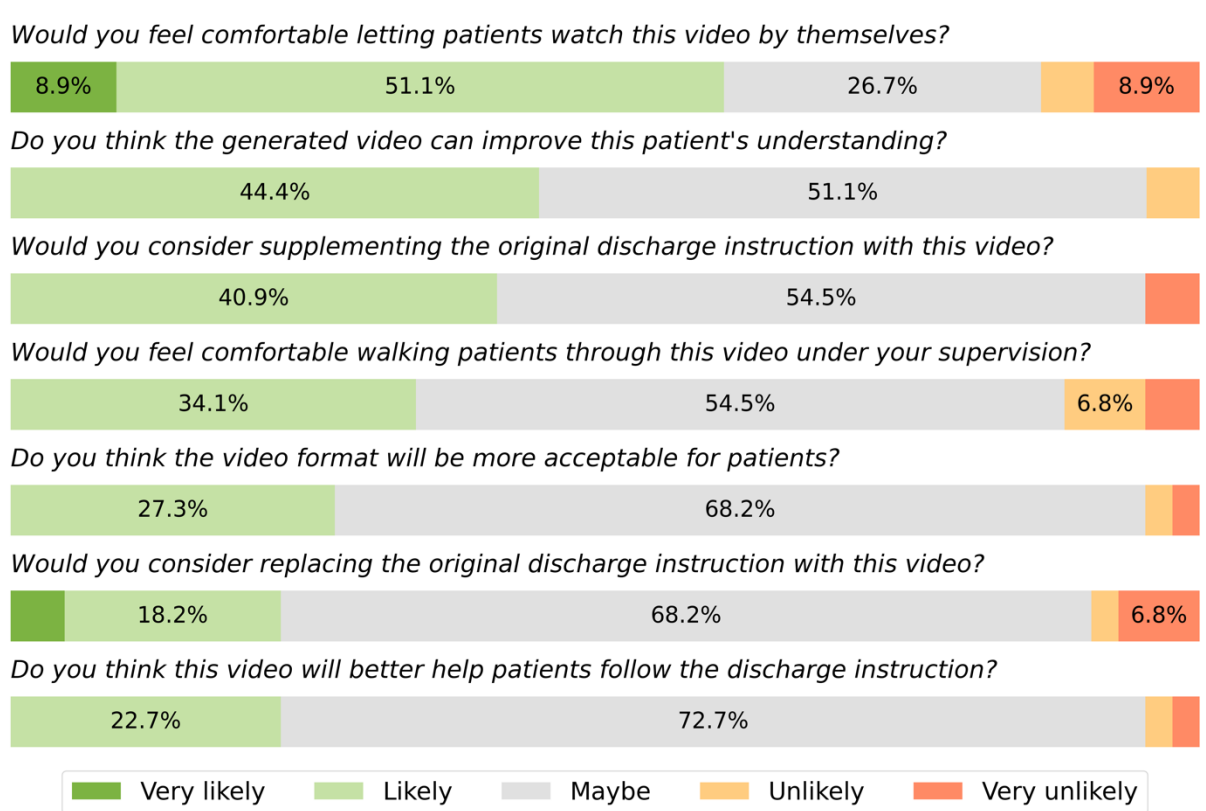
METHOD & CONTRIBUTION

- **Multimodal Integration**: Structured EHR data (demographics, vitals, labs, medications) with unstructured text (radiology impressions, notes).
- **AI Translation:** o3-mini transforms complex medical data into patient-friendly narratives.
- **Video Generation:** Automated conversion to personalized videos featuring virtual healthcare provider using HeyGen.
- **Clinical Evaluation:** Four board-certified Emergency physicians rated 39 discharge instruction pairs on completeness, correctness, and accessibility.



SCAN TO WATCH

DISCUSSION: CLINICAL IMPLICATIONS

- **Physician Endorsement:** 95% of cases rated suitable for supplementing standard instructions; potential to improve patient understanding and adherence.
- **Implementation Readiness:** System requires clinical oversight but provides foundation for scalable discharge communication.
- **Future Directions:** Patient-centered trials needed to assess comprehension, satisfaction, and clinical outcomes; ablation studies to isolate video vs. text benefits.